

01 App Summary

LeadEngine is a white-label, multi-client lead capture CRM built for Novenworks — a marketing agency that runs lead generation for multiple local service businesses. Client websites embed a contact form snippet that routes leads directly into one central platform. Business owners each log in to a private dashboard showing only their own leads. Agency admins see and manage everything across every client.

02 Main User Flows

Agency Admin (Novenworks)

- %, Logs in !' views all leads across all clients on one dashboard
- %, Creates a workspace per client !' generates embed code for their website
- %, Monitors full pipeline, switches between clients, manages accounts

Business Owner

- %, Logs in with business email !' sees only their own leads (enforced server-side)
- %, Moves leads through stages: New !' Contacted !' Booked !' Won / Lost
- %, Adds notes, tracks pipeline value, sees follow-up reminders

Website Visitor (form submitter)

- %, Fills out the embedded contact form on a client's website
- %, Form POSTs to the public /api/leads/capture endpoint
- %, Lead is saved to the database and the business owner receives an email alert

03 Pages & Routes

Route	Access	Purpose
/login	Public	Email + password sign-in
/dashboard	All users	Stats cards, recent leads, follow-up list
/leads	All users	Searchable, filterable leads table (role-scoped)
/pipeline	All users	Drag-and-drop Kanban board by status
/clients	Admin only	All workspace accounts, create/edit/delete
/clients/:id	Admin only	Single workspace detail, embed code, API key
/settings	All users	Profile page (minimal content currently)

Database Models / Tables

users

id, email, name, passwordHash, role (admin|owner), clientId, createdAt

clients

id, businessName, slug, ownerEmail, notificationEmail, apiKey, websiteUrl, industry, isActive, automationEnabled, emailSequenceEnabled, smsSequenceEnabled, aiFollowupEnabled, reviewRequestEnabled, createdAt

leads

id, clientId, name, email, phone, serviceInterest, message, source, status (New|Contacted|Booked|Won|Lost), notes, estimatedValue, monthlyRecurringValue, assignedToId, lastContactedAt, createdAt, updatedAt

activity_log

id, leadId, clientId, userId, action (text), metadata (JSONB), createdAt

05 API Endpoints

Method + Path	Auth	Description
GET /api/healthz	Public	Health check
POST /api/auth/login	Public	Email + password login
POST /api/auth/logout	Session	End session
GET /api/auth/me	Session	Get current user
GET /api/clients	Session	List workspaces (admin=all, owner=own)
POST /api/clients	Admin	Create workspace
PATCH /api/clients/:id	Admin	Update workspace settings/flags
DELETE /api/clients/:id	Admin	Delete workspace
GET /api/clients/:id/embed-code	Session	Get HTML embed snippet + API key
GET /api/leads	Session	List leads with filters
GET /api/leads/:id	Session	Get single lead
PATCH /api/leads/:id	Session	Update status, notes, value, assignment
GET /api/leads/:id/activity	Session	Lead activity log
POST /api/leads/:id/activity	Session	Add activity entry
POST /api/leads/capture	Public (API key)	Submit lead from website form
GET /api/dashboard/stats	Session	Aggregated pipeline stats
GET /api/dashboard/recent	Session	Recent leads feed
GET /api/dashboard/followups	Session	Leads needing follow-up

06 Forms & Data Collected

Login Form

email, password

Public Lead Capture (website embed)

name (required), email (required), phone, serviceInterest, message, source (auto-captured from request)

Create Workspace (admin)

businessName, slug, ownerEmail, notificationEmail, websiteUrl, industry

Lead Update (drawer)

status, notes, estimatedValue, monthlyRecurringValue, lastContactedAt, assignedToId

07 Email / Resend Functionality

%, Library: Resend (requires RESEND_API_KEY environment variable)

%, Trigger: fires on every successful POST /api/leads/capture

%, Recipient: the notificationEmail field set on each client workspace

%, From address: LeadEngine <leads@novenworks.com> — domain must be verified in Resend

%, Content: styled HTML email with name, email, phone, service interest, source, date, message

%, Behavior: fire-and-forget — errors are logged but do not block the API response

%, Missing key: silently skipped with a server-side warning log only

08 Admin vs. Owner Role Differences

Capability	Admin	Owner
View all workspaces (clients)	'	—
Create / edit / delete workspaces	'	—
View leads across all clients	'	—
View own client's leads only	'	'
Update lead status + notes	'	'
Access embed code / API key	'	— (UI only)
Dashboard stats	Global	Own client only

09 What Is Already Working

- %, Full auth lifecycle — login, session persistence, logout, role enforcement
- %, Multi-tenant data isolation — owners cannot see other clients' data (enforced server-side)
- %, Public lead capture with client slug + API key validation
- %, Email notification on every new lead (when RESEND_API_KEY is set)
- %, Dashboard stats with real DB calculations (pipeline value, MRR, follow-up count)
- %, 5-stage lead status pipeline (New !' Contacted !' Booked !' Won !' Lost)
- %, Notes and activity log per lead — every update is timestamped
- %, Drag-and-drop Kanban board (dnd-kit)
- %, Embed code generator per client
- %, Workspace create / edit / delete (admin)
- %, Lead filtering by status, search term, date range, client, and source
- %, Dark premium UI across all pages (global dark mode)

10 What Is Unfinished or Broken

- %, Automation flags (automationEnabled, emailSequenceEnabled, aiFollowupEnabled, etc.) exist in the DB schema but have no backend logic — nothing reads them yet
- %, SMS sequence field (smsSequenceEnabled) exists but no SMS provider is integrated
- %, Lead assignment UI is missing — assignedToId is stored and returned by the API but there is no dropdown to

assign leads in the interface

- % Owners cannot see their own embed code — /clients/:id is admin-only in the frontend router
- % Settings page exists but contains no editable fields (no password change, no profile edit)
- % No way to create owner accounts via the UI — must insert rows directly into the database
- % No lead deletion — leads can be marked Lost but never removed
- % No pagination — the leads list loads all records at once (will slow down at scale)

11 Recommended Next Features

High Impact, Straightforward

- % User management UI — admin can invite/create owner accounts and link to a client
- % Owner embed code access — owners can view their API key and embed snippet from settings
- % Password change form on the settings page
- % CSV export of filtered lead lists
- % Pagination or infinite scroll on the leads list

Medium Impact

- % Email sequence basics — a simple 'send a follow-up email X days after capture' would activate the emailSequenceEnabled flag already in the schema
- % Lead assignment UI — dropdown to assign a lead to a named team member
- % Per-client reporting — monthly volume, conversion rate, source breakdown charts
- % Bulk lead actions — select multiple leads, change status or assign in one click

Longer Term (schema already stubbed)

- % SMS notifications (smsSequenceEnabled field ready — needs Twilio or similar)
- % AI follow-up suggestions (aiFollowupEnabled flag ready)
- % Review request automation (reviewRequestEnabled flag ready)

12 Product Positioning

LeadEngine is the operations layer for local service business agencies. If you run marketing or lead generation for multiple businesses — landscapers, salons, home services, med spas — LeadEngine replaces the chaos of forwarding form emails to separate inboxes.

One platform. Every client. Zero leads lost.

